

COURSE MATERIAL

ENGINEER-IN-SOCIETY
(MNE 201)

FOR

SCHOOL OF ENGINEERING AND ENGINEERING TECHNOLOGY
FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
ONDO STATE, NIGERIA

PREPARED BY
ENGR. (DR) O.C. ALABA

COURSE OUTLINE

- **Section A:** Early Development of Technology
- **Section B:** Science, Engineering and Technology Relationship
- **Section C:** Development of Engineering Profession
- **Section D:** Engineering Regulation and Professional Bodies in Nigeria
- **Section E:** Engineering Education / Training
- **Section F:** Roles and Challenges of Engineering in National Building
- **Section G:** Health, Safety & Environmental (HSE) Management In Engineering Practice
- **Section H:** Engineering Risk Analysis

SECTION A

EARLY DEVELOPMENT OF TECHNOLOGY

➤ What is Technology? Technology:

- ❖ Can be techniques of doing things or
- ❖ Methods by which man solves his environmental problems

➤ History of Technology

- ❖ Started from the creation of man and resides in Garden of Eden to maintain the environment.
- ❖ Records mentioned that the occupation of man during this periods are agriculture, shepherding, hunting.
- ❖ It means man uses some tools and techniques to practices their occupation during this period.
- ❖ This period was described as ***prehistoric years or age*** i.e. earlier than 4000 years (4000 B.C)

➤ Prehistoric Period

- ❖ No clear records on technology development for 4000 BC
- ❖ Finding from prehistoric times was achieved by examining tools and buried objects that were found by Archaeologist
- ❖ Other sources of information include:
 - ✓ Articles made of stone, wooden objects, bones and leather from linen cloth, metal and baskets
 - ✓ Engraving on stones, ornament and paintings found in tombs, caves and other sites that have explored by Archaeologist
 - ✓ Building of Noah's ark with wood, Building of Tower of babel with burnt bricks.
- ❖ All these show the early development of technology by man in prehistoric period.
- ❖ The words “***the stone age, the bronze age and the iron age***” refer to various period relevant to the historic development of technology.
- ❖ The words were used for materials for making tools and weapons by ancient man at different period of times.

➤ **Stone Age**

- ❖ This was the period when man used rock for defense, shelter and other construction.
- ❖ Stone was the first material used along with wood by man. This was supplemented with animal bones, ivory skin and gut
- ❖ Early man used stone to make flint for cutting and shaping other materials such as knives, scrapers, axe and saw.
- ❖ Stone was also used as grinding machine, bow drill bits and arrows for hunting, building bridges, making fire

➤ **Bronze Age**

- This was the period when bronze was used in making all weapons, tools and domestic utilities.
- Bronze is a mixture of tin & copper and It was first hard model material to be used by man.
- The bronze Age led to the introduction of tongs for handling small pieces of metals, development of casting process and introduction of bellows for metallurgical processes.

➤ Iron Age

- ❖ This was the period when iron was used in making all weapons, tools and domestic utilities.
- ❖ Iron Age began about 1300 B.C and its discovery is attributed to the Hittites who lived in Asia Minor.
- ❖ The major metallurgical crissons of the Iron Age blacksmith were the discoveries of the following three processes.
- ❖ Steel-making by carburizing i.e. repeated heating of the iron bullet in contact with charcoal
- ❖ Hardening of steel by sudden sequencing from a high temperature, and
- ❖ Tempering of hardened steel by slow heating temperate temperature to avoid brittleness

➤ **Early Contributors to Technology Development**

❖ **The Egyptians Empire (3000 B.C – 500 B.C)**

- ✓ Developed the use of clay tables for writing; originate decimal and algebraic system; developed standards of length and weight; using stones to make tools and weapons and construction materials etc.

❖ **The Greek Empire (600 B.C and 400 A.D)**

- ✓ This period witness mathematical development, scientific discovery and machinery invention.
- ✓ They developed mechanical equipment; used wood for heating, smelting and cooking; used wood for construction materials (building, furniture, machines etc.)
- ✓ Made use of iron and steel for tools and weapons; invented drawing compasses; developed wheels and bearings

❖ **The Romans Civilization Period**

- ✓ They contributed to the development of technology.
- ✓ Wood was used for architectural constructions, building and bridges.
- ✓ Wrought iron and lead were produced for sheet metals and roofing materials

➤ **History of Materials Discoveries and Uses**

- ✓ The selection of materials through technology and engineering have immensely contributed to the civilization of people and many nations.

❖ **Discovery of fire**

- ✓ Man was able to discover fire in order to solve the problem of lighting.
- ✓ Fire was made by rubbing or striking stones which help man to cook and preserve the food, protect himself from cold and wild beast

❖ **Stone and Wood**

- ✓ Stone and wood occurs naturally and used for the benefit of man kind through technology and engineering development.
- ✓ Stone was used for cutting and sharpening objects while wood was used for building, cooking etc.

❖ **Uses of Animals**

- ✓ Animals were created for the benefit of mankind.
- ✓ Through technology and engineering animal were used by our fore father for transportation, food (meat), wears etc.

➤ History of Materials Discoveries and Uses Cont'd

❖ Solid Minerals

- ✓ Gold is the first metal to be discovered by man and cast, hammered into thin sheets and bowl, draw into wire and joined together by soldering
- ✓ Other minerals discovered by man after gold are copper, bronze and iron which were used for construction of statutes, chisel and spares.
- ✓ Casting of copper give birth to bronze which were used to make tools and weapons for about 2000 years in Egypt.
- ✓ Iron was discovered in 120BC and the discovery is generally attributed to Hittites.
- ✓ Hardening and tempering the brass produced by the cementation process was known as **white bronze**.
- ✓ The Romans used brass for coinage, brass ornaments and fittings
- ✓ Quarry of stone (Granite, limestone etc.) with fire and big hammered were used for building constriction and ornament designs

➤ **Revolution of the Engineering Industries**

- ❖ Took off in United Kingdom in the eighteenth century, spreading to Europe, North America and the world, replacing muscle by machine in a synergistic combination between knowledge and capital.
- ❖ The first Industrial Revolution took place from 1750–1850 and focused on the textile industry.
- ❖ The second Industrial Revolution focused on steam and the railways from 1850–1900
- ❖ The third Industrial Revolution was based on steel, electricity and heavy engineering from 1875–1925.
- ❖ The fourth Industrial Revolution based on oil, the automobile and mass production, taking place between 1900–1950 and onward
- ❖ The fifth phase was based on information and telecommunications and the post-war boom from 1950.
- ❖ These waves of innovation and industrial development have become known as Kondratiev waves, K-waves, long waves, supercycles or surges.

➤ History of Engineering in the World

- ❖ The exact origin of the word 'engineering' comes from the era when humans applied themselves to skillful inventions.
- ❖ The word engineer has its root in the word engine, which comes from the Latin word ingenium, which means "innate quality particularly of mental power".
- ❖ And thus the word engineer emerged as a person who creates well-designed and practical inventions.
- ❖ The exact origin of the engineering profession can be considered to have been at the time the first man used his ingenuity to adapt the materials and forces of nature to meet his needs.
- ❖ Civilization brought miracles into science in the third, fourth and fifth centuries B.C.
- ❖ These were the times when plato (427-347 BC), Aristotle(384-322BC) and Archimedes (287-212 B.C) lived.
- ❖ The modern Science actually came into existence in the 17th century.

- ❖ This century exhibited the works of some of the following scientists and engineers
- ✓ Kepler(1571-1630) laid down the three laws of planetary motion in 1609-1617.
 - ✓ Newton(1642-1727) formulated the laws of motions
 - ✓ Galileo in 1593 invented the thermometer and 1609 use his telescope to discover Jupiter's moons.
 - ✓ John Napier developed Logarithms to aid calculation work. This Lead to the invention of a calculating device called slide Rule.
 - ✓ Leibnitz developed a mechanical computer in 1673.
 - ✓ Charles Babbage an English man started in 1823 the idea of the modern computer and thereafter invented the first automatic machine capable of doing arithmetic at a fast rate.
 - ✓ Alessandro antonio Volta (1745-1827) an italian physicist invented the electric battery in 1796, which was named after him.
 - ✓ Andre-Marie Ampere (1775-1836), a french Mathematician and physicist laid the foundation of electrodynamics.

- ✓ Micheal Faraday (1791-1867), an English chemist and physicist was the greatest experimentalist who ever lived. He discovered the electromagnetic induction in 1831, which opened the door for the generation of electricity.
- ✓ Joseph Henry (1797-1878), an American Physicist, discovered the inductance and constructed an electric motor.
- ✓ Gauss and Weber in 1833 discovered a telegraph apparatus and the morse code.
- ✓ Draisine, a German in 1816, designed the first two-wheeled vehicle called bicycle.
- ✓ Charles and Frank Duryea made the first American internal combustion motor car in 1893.
- ❖ The list of Scientist and Engineers is endless, man is creating, solving problems on a daily basis and has moved into other planets for researches.
- ❖ The rapid advance of technology has brought forth many societies and many technological and professional organizations of potential interest to Engineers such as the World Federation Of Engineering Organization (WFEO).
- ❖ The establishment of this unity organization has brought about the further development of the engineering profession.

➤ History of Engineering in Nigeria

- ❖ The history of engineering in Nigeria traces its root to the establishment of the Public Works Department (P.W.D) of southern Government in Lagos in 1896.
- ❖ The P.W.D comprised mainly of three sections (Civil, Mechanical and Electrical).
- ❖ It was responsible for the management of engineering problems in the southern Government of Nigeria.
- ❖ As the country entered into independence in 1960, the three regions (North, East and West) had their separate P.W.D.
- ❖ In 1966, the Nigerian council for scientific and industrial Research (NCSIR) was established. The NSCIR gave rise to the establishment of:
 - ✓ International Institute of Tropical Agriculture, Ibadan (IITA) in 1967,
 - ✓ Kaduna polytechnic in 1968,
 - ✓ Yaba College of technology in 1969, and a
 - ✓ New council called Nigerian Council For Science and Technology (NCST).
- ❖ The creation of the NCST produced the promulgation of Nigerian Steel Development authority (NSDA) Decree of 1971.

- ❖ In the same year, the Agricultural Research Council of Nigeria (ARCN) was established.
- ❖ Also , in 1971 on the recommendation of NCST, the industrial Training Fund (ITF) was established.
- ❖ In 1973, the Petroleum Technology Development Fund was created to train Nigerians in the field of Engineering, Geology, Science and Management in the petroleum Industry.
- ❖ This lead to the establishment of the Petroleum Training Institute, Warri.
- ❖ The National Science and Technology Development Agency (NSTDA) was established in 1977 and in 1980, a separate Ministry of science and Technology was born.
- ❖ In compliance with Vienna(1979) ,directives and recommendations, apart from the existing specialized Federal Polytechnics, more Federal Universities of Technology were also established in 1980 to cater for the training of engineering manpower .
- ❖ In 1987, the National Science and Technology Fund (NSTF) was put in place to provide fund for activities in science and technology

- ❖ In 1988, the Raw Materials Research and Development Council (RMRDC) was born for the development of industrial raw materials, creating self sufficiency and maximization of inputs to local industries.
- ❖ Evidence of Engineering and Technology in Nigeria is seen in development and adaptation of appropriate machines and equipment for easing out tediousness and labour in certain operations in agriculture and Industries developed by these government agencies.
- ❖ These indigenously developed adapted machines and equipment are now becoming regular exhibits at trade fairs ,shows and exhibition that are regularly organized by government agencies, associations and private industrialists.
- ❖ In the development of engineering profession in Nigeria, regulation has been carried out by the professionals themselves, through learned societies and a statutory registration and regulatory body.
- ❖ The learned societies and a statutory registration and regulatory body are:
 - ✓ Nigeria Society of Engineers (NSE),
 - ✓ Council for the Regulation of Engineering in Nigeria (COREN)

SECTION B

HISTORICAL DEVELOPMENT OF SCIENCE, ENGINEERING AND TECHNOLOGY

- Below are the basic relationship between Science, Engineering and Technology

Science	Engineering	Technology
Science uses different approaches or scientific methods, like longitudinal observational studies or controlled experiments which helps to produce learning.	Engineering uses different approaches like, building design processes or designing examinations which helps to create and assess arrangements and technologies.	Technology is actually the results from the procedure of engineering
The knowledge of science can be utilized to construct predictions.	Engineering can be used to deliver the best arrangements given constraints and resources	Technologies are anything made by people and it is used to fulfill a need or wants of people.
The example of Science is burning wood that produces water, heat and carbon dioxide	The example of engineering is to build a chimney and fireplace makes it simpler to cook with flame without filling the smoke in room	The example of technology is that fire can be utilized to cook different types of food.

Science, Engineering and Technology Relationship Cont'd

Figure 1: Relationship among Science, Technology and Engineering

SECTION C

Development of Engineering Profession

➤ Engineering Profession

- ❖ The engineering profession is defined as “the collection of man-made or induced physical objects located in a particular area or region.
- ❖ It creates the physical world that has been intentionally created through science and technology for the benefit of mankind.
- ❖ The purpose of regulating a profession is ‘to assure the quality of professional services in the public interest.
- ❖ The regulation of a profession involves:
 - ✓ the setting of standards of professional qualifications and practice;
 - ✓ the keeping of a register of qualified persons and the award of titles;
 - ✓ determining the conduct of registrants,
 - ✓ the investigation of complaints and disciplinary sanctions for professional misconduct.

- ❖ There are a number of approaches to the regulation of a profession around the world. Broadly speaking, these include:
 - ✓ Licensing: to authorize eligible persons to practise in a specific area
 - ✓ Registration: to recognize demonstrated achievement of a defined standard of competency
 - ✓ Specialist lists: to indicate peer-recognized competence in a particular area
- ❖ All these forms of regulation are linked to codes of conduct.
- ❖ Serious breaches of a code of conduct can lead to:
 - ✓ the withdrawal of a license,
 - ✓ the loss of a title or the removal of the transgressor's name from a specialist list, either on a temporary or permanent basis.

➤ Codes of Practice and Standards

- ❖ Generally, codes and standards are dependent on the environmental factors such as humidity, temperature, pressure and atmospheric particle density prevailing in different country of origin

- ❖ Codes and standards for engineering design and construction are used as part of training materials for prospective engineers.
- ❖ In Nigeria, we rely on codes and standards from British Standards Institute and American National Standards Institute both for the training and professional practice.
- ❖ The Standards Organisation of Nigeria has only been successful at providing standards for quality of some commercial products but not for engineering projects.

➤ Ethics in Engineering

- ❖ Engineers shall hold paramount the Health, Safety and Environment/welfare of the public in the practice of their profession.
- ❖ Engineers shall practice only in their areas or field of competence, in a careful and diligent manner and in conformance with standards, laws, codes, rules and regulations applicable to engineering practice.
- ❖ Engineers shall examine the societal and environmental impact of their actions and projects, including the use and conservation of resources and energy in order to make informed recommendations and decisions..
- ❖ Engineers should declare their interests clearly.

- ❖ Engineers shall sign and take responsibility for all engineering works which they prepared or directly supervised. Engineers may sign works prepared by others only with their consent and after sufficient review and verification.
- ❖ Engineers shall act as faithful agents for their employers or clients and maintain confidentiality, avoid conflicts of interest whenever possible and disclose unavoidable conflicts.
- ❖ Engineers professional concerns must be made known to the client and the consequences of engineering decisions or judgments.
- ❖ Engineers should reject any public works, engineering decisions or practice that endanger the HSE of the public.
- ❖ Engineers shall commit to life-long learning, strive to advance the body of engineering knowledge and should encourage other engineers to do likewise.
- ❖ Engineers shall promote responsibility, commitment and ethics both in the education and practice phases of engineering.
- ❖ They should enhance society's awareness of engineer's responsibilities to the public and encourage the communication of these principles of ethical conduct among engineers

SECTION D

Engineering Regulation and Professional Bodies in Nigeria

- Council for the Regulation of Engineering Practices in Nigeria (COREN)
 - ❖ COREN is the only engineering body established by Federal Government and the highest Engineering council in Nigeria
 - ❖ It was established in 1970 by decree No. 55 with the name Council for Registered Engineers in Nigeria (COREN)
 - ❖ The Decree establishing COREN was amended by Decree 27 of 1992, now the “**Engineers Act, CAP E110 of 2004**” Law of the Federal Republic of Nigeria.
 - ❖ The name Council for Registered Engineers in Nigeria was changed to Council for the Regulation of Engineering Practices in Nigeria (COREN).
 - ❖ The Decree and Act state that: There shall be established on the coming into force of this decree, body to be known as the council of Registered Engineers of Nigeria (hereafter in the decree referred to a “council which shall be a body corporate by the name afore said and charge with the general duty of: **24**

- ✓ Determining who are engineers for the purpose of this Decree;
- ✓ Determining what standards of the knowledge and skill are to be attained by person seeking to become registered as Engineers and raising these standards from time to time as circumstances may permit;
- ✓ Securing in accordance with the provisions of this Decree the establishment and maintenance of a register of persons entitled to practice as registered engineers and the publication from time to time of lists of those persons.
- ✓ Performing the functions conferred on the council by this Decree;
- ✓ Regulating and controlling the practice of the engineering profession in all its aspects and ramifications.
- ✓ Others responsibilities of the Council are:
 - Registration of engineers
 - Approval of courses qualification and accreditation of institutions for Engineering programme
 - Supervision of instruction and examinations leading to approval qualifications;
 - Maintain professional discipline among members

- ❖ COREN has identified four cadres in the practices of Engineering in Nigeria.
- ❖ These are the engineer, the technologist, the technician and craftsman.
- ❖ The British engineering Council (1920) defines each cadre as follow:

➤ The Engineer

- ❖ An engineer is one who requires and uses scientific, technical and other pertinent knowledge and skill to create or maintain safe, efficient system, structures, machines, plant, processes and devices of practical and economic value.
- ❖ Engineers are problem solvers, creators of ideas and concepts, builders of devices, structures, and systems.
- ❖ Roles of Engineer
 - ✓ He shall has the ability to think in abstract based on his educational training
 - ✓ He is involved in preparing plans and teaching of engineering courses
 - ✓ He must involve in research, design, development of projects and in operation and maintenance of complex engineering plant services and establishments

➤ The Engineering Technologist

- ❖ They apply engineering and scientific knowledge combined with technical skills to support engineering activities.
- ❖ As a group, they are important problem solving individuals whose interests are directed more to the practical than to the theoretical aspect of a project.
- ❖ Roles of Engineering Technologist
 - ✓ He performs technical duties of an establishment either independently or under the supervision of registered engineer or scientist
 - ✓ He present techniques in the execution and management of engineering works
 - ✓ He assists the engineer in the design and development of engineering plant and structures where appropriate.

➤ The Engineering Technician

- ❖ Work with equipment, primarily assembling and testing component parts of devices or systems that have been designed by others; usually under direct supervision of an engineer or engineering technologist.

- ❖ Their preferences are given to assembly, repair, or to making improvements to technical equipment by learning its characteristics, rather than by studying the scientific or engineering basis for its original design.
- ❖ They may carry out standard calculations, serve as technical sales people, make estimates of cost, assist in preparing service manuals, or perform design-drafting activities.
- ❖ They are frequently employed in laboratories and/or manufacturing facilities where they may set up experiments, accumulate scientific or engineering data, and/or service or repair engineering or production equipment.

➤ Craftsman

- ❖ Programs of study are also available for individuals who wish to obtain skill-training in a field of specialization with less emphasis on scientific or mathematical principles.
- ❖ An individual completing such a program is typically called a "specialized" technician, e.g., air-conditioning technician, draftsman, surveyor aide, etc

➤ Engineering Registration

- ❖ Engineering registration is a way of protecting the rights, health and property of the public and the profession
- ❖ Legal registration of engineers are met for those who have required qualifications and can safely serve the public
- ❖ This is achieved by prescribing minimum standard for registration for various cadres of engineering profession as stated below:
 - ✓ B. Eng. degree or equivalent certificate is required for Engineer
 - ✓ HND degree or equivalent certificate is required for Technologist
 - ✓ ND degree of or equivalent certificate is required for Technician
 - ✓ O' Level or equivalent certificate is required for Craftsman
- ❖ In addition, the candidate must have undergone training and acquired experiences under the supervision of registered engineer for a minimum of 2years
- ❖ COREN is the only legal body that can register and give license to practice engineering in whatever form in Nigeria.

➤ Importance of Engineering Registration

❖ To the Public

- ✓ Registration protecting the public from quacks, chalanants and unworthy engineers;
- ✓ It provides a strict yard-stick against which various cadres of engineering profession to be judge on their ability to do specific jobs

❖ To the Employer

- ✓ Save time when recruiting through the use of identified internationally;
- ✓ Show investors, analysts and customers of well qualified staff;
- ✓ The employer uses staff with qualifications recognized internationally;
- ✓ Demonstrates that the aspirations of new employees can be satisfied;
- ✓ Registration assures a skill base to cope with new technologist and make change; and
- ✓ Assures customers of professional standards to meet legal requirement

❖ To the Individual

- ✓ Registration increases mobility across wide range of employment;
- ✓ Creates opportunities for satisfying work and tangible results for the individual;
- ✓ Provides qualification of international standing known and understood by employers and their customers;
- ✓ Keeps individual abreast of latest development;
- ✓ Provides proof of standard of education, training and experience; and
- ✓ Stimulates career development

➤ The Nigerian Society of Engineers (NSE)

- ❖ Founded in 1958, and NSE is the national umbrella organization for the Engineering Profession in the country.
- ❖ Is a body dedicated to the promotion of engineering knowledge and its practical application to members
- ❖ The major objective of the Society is to promote the advancement of engineering education, research and practice in all its ramifications.

- ❖ Membership is open to all who possess qualifications that are acceptable by Council for the Regulation of Engineering in Nigeria (COREN).
- ❖ Commencing in 1988, the society conducted its own Graduate ship Examinations for candidates who are not otherwise qualified.
- ❖ There are six grades of membership, namely: - Student, Graduate, Corporate, Associate, Fellow and Honorary Fellow.
- ❖ Student Member
 - ✓ Student member shall be a person who is undertaking an approved engineering course of training leading to degree or diploma award in an institution recognized by the council of the society for the purpose of training professional engineers
- ❖ Graduate Member
 - ✓ A graduate membership shall possess an academic qualification acceptable to the council of the society and registrable by the COREN under the provisions of Engineers Registration Decree (No. 55) of 1970 or such amendments thereof.
 - ✓ Post-qualification practical experience shall not be requirement for entry into this class

❖ Corporate Member

Category A

- ✓ Candidate who are holders of the bachelor's degree in Engineering whose institution programmes are accredited by COREN and with such candidates having registered by COREN as professional Engineers
- ✓ Candidate who is already corporate members of engineering institution registered by COREN

Category B

- ✓ Candidate who are holders of the bachelor's degree in Engineering whose qualification are accredited by COREN or any overseas accreditation Boards recognized by COREN. Such candidate must have acquire a minimum of 4 years post qualification experience.
- ✓ Candidate who are graduate of unaccredited engineering courses from Nigerian Universities and required a continuous 6 year post graduate experience in the interview on their personal merit
- ✓ Candidates who have obtained full pass in the NSE Graduateship Examination and have had additional 2 years of engineering practice

- ✓ Candidates who obtained HND in engineering before proceeding to do a master degree or Post Higher National Diploma in engineering

Category C

- ✓ Candidate who are holders of correct and adequate basic academic qualification by COREN standards and regulations who are 40 years and above with minimum of 15 years post qualification experience.

❖ Associate Member

- ✓ An associate membership shall possess an academic qualification of university degree in the science allied to engineering science or other qualification approved by the council of the society. In addition, shall have been engaged on work related to the practices of engineering for a minimum period of 5 years

❖ Fellow Member

Election to fellowship of the society shall be either by invitation of the executive council of the society or by direct application by a member

- ✓ Candidate to be invited to fellowship of the society shall have not less than 15 years experience in the practice of engineering science and shall have satisfied the requirements for corporate membership of the society

- ❑ In addition, he shall have met one or all of the following conditions:
 - Have rendered valuable and selfless services to the society over a period who a period time
 - Have been outstanding in field of engineering
- ✓ A candidate seeking fellowship of the society by application shall have had not less than 10 years experience in the practice of engineering science and shall have satisfied the requirements for membership
- ❑ In addition, he shall meet one or all of the following conditions:
 - Submit a dissertation containing valuable statistics on engineering practice in Nigeria
 - Submit a dissertation or thesis on original solution to a particular engineering problem in Nigeria
 - Submit dissertation adjudged by council to be a worthwhile contribution to engineering knowledge in Nigeria

➤ Honorary Fellows

- ❖ The council of the society may at his discretion to elect distinguish persons to Honorary fellowship.

- ❖ Such person is not necessarily be engineers but would normally have their position to render notable assistance to the practice of engineering science
- Note:
 - ❖ Only a corporate members and fellows are entitled to vote and voted for at properly constituted meetings of the society
 - ❖ A person seeking to join the NSE shall fill the application form obtain from either the society head/branches offices or downloaded from society website
 - ❖ A completed application must be submitted which must be sponsored by two corporate or fellow financial members
 - ❖ All candidates shall be tested orally. In addition, candidates except those in category C shall also take compulsory written tests
 - ❖ Category B candidates shall present reports on the their post qualification/graduation experiences and write essay during their interviews

➤ Division of NSE

❖ Some of the engineering division under NSE are:

- ✓ The Nigerian Society of Engineers;
- ✓ The Nigerian Society of Mining Engineers;
- ✓ The Nigerian Society of Chemical Engineers;
- ✓ The Nigerian Society of Agricultural Engineers
- ✓ The Nigerian Society of Production Engineers
- ✓ The Nigerian Association of Metallurgy
- ✓ Foundry Association of Nigeria
- ✓ The Nigerian Association of Technological Engineers
- ✓ Energy Society of Nigeria
- ✓ The Nigerian Society of Production Engineers

➤ The Council of Nigerian Mining Engineers and Geoscientists (COMEG)

- ✓ COMEG was established in 2003 by government to provide registration for Geoscientists and Mining Engineers in Nigeria

SECTION E

Transfer of Technology

➤ **Meaning of Transfer of Technology**

- ❖ Transfer of technology involves borrowing of technical skill from a country
- ❖ Organization of American State (OAS) defined transfer of technology as the process through which the production system of a country acquires the technology produced in another country for incorporation in their businesses
- ❖ Federal Ministry of Science and Technology (FMST) defined transfer of technology as adoption of technological skill or knowledge based on the experience of others through studying or utilization of a device to produce similar device.

SECTION F

Roles and Challenges of Engineering in National Building

➤ Roles of Engineering in National Building

❖ Engineers play a critical role in national building which includes:

- ✓ **Agricultural sector:** Engineering has really promoted the practice of agriculture all over the world e.g. production of fertilizers by chemical engineers, provision of water for irrigation by civil or mining engineers, construction of machine for ploughing, harrowing etc.
- ✓ **Banking sector:** Engineering plays fundamental roles in banking institutions of various societies through provision of software and networking (software and networking), maintenance of computer systems (computer engineers) and manufacturing of money counting machines (mechanical engineers). Construction and testing of durable bullet proof doors (Metallurgical and materials engineers).
- ✓ **Automobile:** Design and construction of automobiles (mechanical engineering), making the alloys and cast of automobile engine block, piston and the gears (metallurgical engineers),

- ✓ **Educational Sector:** The contributions of engineering in education sector includes provision of clothes for students and teachers (textile engineers), computers, air-conditioners and laboratory equipment (computer and mechanical engineers)
- ✓ **Marketing:** Engineering contributes to the development of marketing activities in our society through provision of internet and website design, computers and mobile phones and graphic designs.
- ✓ **Health Sector:** Engineering contributes to the development of health sector through provision of equipment in diagnosing patients and surgery equipment like scalpel, surgical lamps, stethoscope etc.
- ✓ **Community Development:** Engineering promotes community development through creation, improvement, and protection of the communal environment, providing facilities for living e.g. buildings, roads, bridges, canals, railroad lines, airports, water-supply systems etc.

➤ Challenges of Engineering In National Building

- ❖ Engineers being all in all: this occurs when right engineers were not used in executing projects.

- ❖ Corruption: this involves eating money for engineering capital project rather profit without carrying out or complete the engineering projects.
- ❖ Non-Engineers carrying out engineering contracts using engineering credentials
- ❖ Existing engineering facilities and infrastructures not being upgraded: most engineering establishment ever since they were commissioned have not been upgraded , thereby not able to meet up with the present day demand.
- ❖ Attitude towards maintenance: our industries and infrastructures are built “once and for all” without any routine maintenance work which general decay of industries and infrastructures in the country
- ❖ Reverse engineering not in our engineering curriculum; hence, making technology transfer some how difficult.
- ❖ Engineers not fellowshipping with their colleagues: most engineering personnel occupying managerial positions are not registered with the NSE and COREN, thereby pay little or attention for the advancement of engineering and technology in the country
- ❖ Research and Development: Government lackadaisical attitude towards research and development hinders research opportunities in research institutions and universities.

➤ Solution to Engineering Challenges In Nigeria

- ❖ Employment of right engineers in their areas of specialization
 - ❖ The pay package of engineers in Nigeria should be commensurate with their counterparts in Europe and America.
 - ❖ Engineers should be discipline and avoid non-engineers using them to achieve their selfish aim
 - ❖ Just as the office of the Attorney General is occupied by a lawyer, the offices of the ministers and commissioners of energy, works and housing, Environment and transport should be specially for engineers
 - ❖ Existing engineering facilities and infrastructures should be upgraded with the present day state of the art facilities in order to meet up with the present day demand
 - ❖ Routine maintenance work should be carried out, on a regular basis after a project have been commissioned, this will increase the life span of such infrastructures and facilities
 - ❖ Reverse engineering should be introduced into our educational curriculum , in order to make technology transfer very easy
- 42**
-
- Non-Engineers carrying out engineering contracts using engineering credentials
 - Existing engineering facilities and infrastructures not being upgraded: most engineering establishment ever since they were commissioned have not been upgraded , thereby not able to meet up with the present day demand.
 - Attitude towards maintenance: our industries and infrastructures are built “once and for all” without any routine maintenance work which general decay of industries and infrastructures in the country
 - Reverse engineering not in our engineering curriculum; hence, making technology transfer some how difficult.
 - Engineers not fellowshiping with their colleagues: most engineering personnel occupying managerial positions are not registered with the NSE and COREN, thereby pay little or attention for the advancement of engineering and technology in the country
 - Research and Development: Government lackadaisical attitude towards research and development hinders research opportunities in research institutions and universities.

SECTION G

Health, Safety & Environmental (HSE) Management In Engineering Practice

- ❖ Health, Safety and Environmental management should be part of the engineering profession in a country for the purpose of
 - ✓ duty of care
 - ✓ economic reasons and
 - ✓ legal reasons.
- ❖ HSE management involves the following broad phases:
 - ✓ Specifications
 - ✓ Design and implementation
 - ✓ Installation and commissioning
 - ✓ Operation and maintenance
 - ✓ Changes after commissioning.

- ❖ Compliance with the standards requires four essential elements:
 - ✓ Identification of safety functions required for the safe shutdown
 - ✓ Assignment of a safety integrity level (SIL) for each safety function
 - ✓ Use of the safety lifecycle for the engineering design and
 - ✓ Verification of the SIL achieved for each safety function.

➤ Engineering Code of Practice

- ❖ The engineering code of practice takes into consideration the following
 - ❖ Risk Management: Taking reasonable steps to minimize the risk of loss of lives, injuries or suffering.
 - ❖ Workplace and construction site: Minimizing potential dangers involved in the construction and manufacture of engineering products and processes.
 - ❖ Public/Community well-being
 - ❖ Communication
 - ❖ Conflicts of interest
 - ❖ Confidentiality

➤ Safe Engineering Designs

- ❖ Safety is the antonym of risk. So, a design is safe to the extent that it reduces risk.
- ❖ A safe design is the combination of all those procedures and principles that are used by engineers to make designed objects safe against accidents leading to human death or injuries, long term health effects, damage to the environment or malfunctioning in general.
- ❖ Safety engineering principles also include education of operators, maintenance of equipment and installations and incidence reporting are examples of safety practices of general importance.
- ❖ Several design strategies used to achieve safety in operations of potentially dangerous technology are:
 - ✓ Inherently safe design
 - This minimizes the inherent dangers in the process as far as possible.
 - Potential hazards are excluded rather than enclosed or coped with.
 - For instance, dangerous substances are replaced by less dangerous ones and fire proof materials are used rather than inflammable ones.

✓ Safety factors

- Construction should be strong enough to resist load and disturbances exceeding those that are intended.
- A common way to obtain such safety reserves is to employ explicitly chosen numerical safety factors are employed.
- If a safety factor of two (2) is employed when building a bridge, then the bridge is calculated to resist twice the maximal load to which it will be exposed to in practice

✓ Negative feedback (self-shutdown)

- This is introduced to achieve a self-shutdown in case of device failure or when the operator loses control.
- Examples are safety valves that let out steam when the pressure is too high in a steam boiler and the dead man's hole that stops the train when the driver falls asleep.
- One of the most important safety measures in the nuclear industry is to ensure that reactors close down automatically in critical situations.

- ✓ Multiple independent safety barriers.
 - Safety barriers are arranged in chains, so that each barrier is independent of its predecessors (if the first fails, the second is still intact).
 - The first barriers prevent accidents; the second barriers limit the consequences of an accident and rescue services as the last resort.
 - Safety factors and multiple safety barriers deal with uncertainties as well as risks.
 - But currently, Probabilistic Risk Analysis (PRA) is used but does not deal with uncertainties.

➤ Bad Engineering Practices

- ❖ Seven (7) bad engineering practices have been identified:
 - ✓ Believing that if something is not specifically stated, either "shall do" or "shall not do" in the standards, an engineer does not need to worry about it.
 - ✓ Thinking that meeting the minimum requirements means the process is safe and complies with the standard.
 - ✓ Ignoring the importance of good engineering practice.

- ✓ Designing systems that meet economic requirements but not safety protection requirements.
- ✓ Neglecting human factors (errors in calculations etc)
- ✓ Focusing on capital cost and not on lifecycle costs.
- ✓ Focusing only on the safety integrity level (SIL) and not on prevention.
- ❖ In engineering design, safety consideration always includes safety against unintended human death or injuries that occur as a result of the unintended use of the designed object for:
 - ✓ Prevention of damage to the environment
 - ✓ Prevention of long term health effects

➤ HSE Sustainability Management

- ❖ This is about the long term survival of humanity.
- ❖ It recognizes that decisions made today must enable both those in the present as well as people of the foreseeable future to make effective choices about their quality of life.

- ❖ Failure to identify risks to safety and the inability to address or control these risks can result in massive costs, both human and economic.
- ❖ The multidisciplinary nature of safety engineering means that a very broad array of professionals are actively involved in accident prevention or safety engineering.
- ❖ A critical fault endangers of few people.
- ❖ A catastrophic fault endangers, harms or kills a significant number of people.
- ❖ Engineer's errors or inability to incorporate the HSE management in his practice spells catastrophic.

➤ The Way Forward

- ❖ Everyone must strengthen his or her understanding of HSE awareness by making safety a priority.
- ❖ Also, cost effective solutions in order to gain the biggest return on investment should be developed.
- ❖ The following principles should adhere strictly to ensure engineering safety
- ❖ At all times, take all reasonable care to ensure that your work and the consequences of your work cause no unacceptable risk to safety.

- ❖ Take all reasonable steps to make your management/client and those to whom they have a duty of care aware of the risks you identify.
- ❖ Make anyone overruling or neglecting your professional advice formally aware of the consequent risks.
- ❖ It is critical for engineers to maintain a deep and broad understanding of the many technical and professional practice issues that they will inevitably encounter in their role as employees of public owners.
- ❖ This is achieved through appropriate education, training, experience, license, professional engineering practice and continuing professional development.

SECTION H

Engineering Risk Analysis

➤ Risk management

- ❖ Risk is an event that, if it occurs, adversely affects the ability of a project to achieve its outcome objectives
- ❖ Risk management is the process of identifying risk, assessing risk, and taking steps to reduce risk to an acceptable level
- ❖ The risk management approach determines the processes, techniques, tools, and team roles and responsibilities for a specific project.
- ❖ The risk management plan describes how risk management will be structured and performed on the project
- ❖ Historically, risk management focused more on management elements such as schedule and cost, and less on technical risks for well-defined or smaller projects.
- ❖ To increase the likelihood of successful project and program outcomes, the systems engineer and project manager must be actively involved in all aspects of risk management

➤ Steps in Engineering Risk Analysis

❖ **Step 1: Risk identification:**

- ✓ It is the process of determining risks that could potentially prevent the program, enterprise, or investment from achieving its objectives.
- ✓ It includes documenting and communicating the concern
- ✓ It is the critical first step of the risk management process see Figure 3
- ✓ The objective of risk identification is the early and continuous identification of events that, if they occur, will have negative impacts on the project's ability to achieve performance or capability outcome goals.
- ✓ They may come from within the project or from external sources.

❖ **Step 2. Risk Impact or Consequence Assessment**

- ✓ In this step, an assessment is made of the impact each risk event could have on the engineering system project.
- ✓ Typically, this includes how the event could impact cost, schedule, or technical performance objectives. Impacts are not limited to only these criteria.

- ✓ The results of this assessment are then used to prioritize risks to establish a most-to-least-critical importance ranking.

❖ **Step 3. Risk Prioritization**

- ✓ At this step, the overall set of identified risk events, their impact assessments, and their occurrence probabilities are "processed" to derive a most critical to least critical rank-order of identified risks.
- ✓ Ranking risks in terms of their criticality or importance provides insights to the project's management on where resources may be needed to manage or mitigate the realization of high probability/high consequence risk events.
- ✓ When assessing risk, it is important to match the assessment impact to the decision framework.
- ✓ For program management, risks are typically assessed against cost, schedule, and technical performance targets as shown Table 1 & 2.

❖ **Step 4. Risk Mitigation Planning, Implementation, and Progress Monitoring**

- ✓ Risk mitigation planning is the process of developing options and actions to enhance opportunities and reduce threats to project objectives.

Table 1. Program Risk Management Assessment Scale Example

1.00	Issue:	1	Certain to occur
0.95-0.99	High:	$> 0.95 < 1$	Extremely sure to occur
0.85-0.95	High:	$> 0.85 \leq 0.95$	Almost sure to occur
0.75-0.85	High:	$> 0.75 \leq 0.85$	Very likely to occur
0.65-0.75	High:	$> 0.65 \leq 0.75$	Likely to occur
0.55-0.65	Medium:	$> 0.55 \leq 0.65$	Somewhat greater than an even chance
0.45-0.55	Medium:	$> 0.45 \leq 0.55$	An even chance to occur
0.35-0.45	Medium:	$> 0.35 \leq 0.45$	Somewhat less than an even chance
0.25-0.35	Low:	$> 0.25 \leq 0.35$	Not very likely to occur
0.15-0.25	Low:	$> 0.15 \leq 0.25$	Not likely to occur
0.00-0.15	Low:	$> 0.00 \leq 0.15$	Almost sure not to occur

- ✓ The risk mitigation step involves development of mitigation plans designed to manage, eliminate, or reduce risk to an acceptable level.
- ✓ Once a plan is implemented, it is continually monitored to assess its efficacy with the intent of revising the course-of-action if needed.
- ✓ Risk mitigation implementation is the process of executing risk mitigation actions.
- ✓ Risk mitigation progress monitoring includes tracking identified risks, identifying new risks, and evaluating risk process effectiveness throughout the project

➤ Risk Mitigation Strategies

- ❖ General guidelines for applying risk mitigation handling options are shown in Figure 4.
- ❖ These options are based on the assessed combination of the probability of occurrence and severity of the consequence for an identified risk.
- ❖ Risk mitigation handling options include:
 - ✓ Assume/Accept: Acknowledge the existence of a particular risk, and make a deliberate decision to accept it without engaging in special efforts to control it.

- These guidelines are appropriate for many, but not all, projects and programs.

Figure 4. Risk Mitigation Handling Options

- ✓ Avoid: Adjust program requirements or constraints to eliminate or reduce the risk. This adjustment could be accommodated by a change in funding, schedule, or technical requirements.
- ✓ Control: Implement actions to minimize the impact or likelihood of the risk.
- ✓ Transfer: Reassign organizational accountability, responsibility, and authority to another stakeholder willing to accept the risk.
- ✓ Watch/Monitor: Monitor the environment for changes that affect the nature and/or the impact of the risk.

❖ Risk management tools

- ✓ It is support the implementation and execution of program risk management in systems engineering programs.
- ✓ Risk analysis and management tools serve multiple purposes and come in many shapes and sizes.
- ✓ Some risk analysis and management tools include those used for:
 - Strategic and Capability Risk Analysis•—Focuses on identifying, analyzing, and prioritizing risks to achieve strategic goals, objectives, and capabilities.

- Threat Analysis—•Focuses on identifying, analyzing, and prioritizing threats to minimize their impact on national security.
- Investment and Portfolio Risk Analysis—•Focuses on identifying, analyzing, and prioritizing investments and possible alternatives based on risk.
- Program Risk Management—•Focuses on identifying, analyzing, prioritizing, and managing risks to eliminate or minimize their impact on a program's objectives and probability of success.
- Cost Risk Analysis—Focuses on quantifying how technological and economic risks may affect a system's cost. Applies probability methods to model, measure, and manage risk in the cost of engineering advanced systems.

❖ Selecting the Right Tool

- ✓ It is important that the organization defines the risk analysis and management process before selecting a tool. Ultimately, the tool must support the process.
- ✓ Below are criteria to consider when selecting a risk analysis and management tool:

- Aligned to risk analysis objectives—•Does the tool support the analysis that the organization is trying to accomplish? Is the organization attempting to implement an ongoing risk management process or conduct a one-time risk analysis?
- Supports decision making—•Does the tool provide the necessary information to support decision making?
- Accessibility—•Is the tool accessible to all users and key stakeholders? Can the tool be located/hosted where all necessary personnel can access it?
- Availability of data—•Is data available for the tool's analysis?
- Level of detail—•Is the tool detailed enough to support decision making?
- Integration with other program management/system engineering processes—•Does the tool support integration with other program management/system engineering processes?